Week 19 Self-Assessment

APA Reporting Format

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1 Instructions

Test your understanding of APA format for reporting statistical results. Correct reporting is essential for scientific communication and is assessed in your lab report and exam.

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2 Part 1: Conceptual Questions

2.1 Question 1: Why APA Format?

APA format for statistics is important because:

- (A) It makes papers look professional
- (B) It provides a standardised way to communicate findings that all researchers understand
- (C) Journals require it for publication
- (D) All of the above

2.2 Question 2: General Formatting Rules

In APA format, statistical symbols (like *t*, *p*, *r*, *M*, *SD*) should be:

- (A) Bold
- (B) Italicised
- (C) Underlined
- (D) In quotation marks

How many decimal places are typically reported for means and standard deviations?

- (A) 0
- (B) 1
- (C) 2
- (D) As many as available

2.3 Question 3: Leading Zeros

For values that cannot exceed 1 (like *p*, *r*, and proportions), APA format:

- (A) Always uses a leading zero (0.05)
- (B) Does not use a leading zero (.05)

- (C) Uses both interchangeably
- (D) Uses percent instead

Which is correct for reporting a p-value?

- (A) $p = 0.032$
- (B) $p = .032$
- (C) $P = .032$
- (D) p-value = 0.032

2.4 Question 4: Very Small p-Values

When a p-value is very small (e.g., .00004), the correct way to report it is:

- (A) $p = .00004$
- (B) $p = .000$
- (C) $p < .001$
- (D) p is significant

2.5 Question 5: Degrees of Freedom

Degrees of freedom should be reported:

- (A) At the end of the sentence
- (B) In parentheses after the test statistic symbol
- (C) Only if requested
- (D) In a separate table

2.6 Question 6: Rounding Rules

Values should generally be rounded to:

- (A) Whole numbers
- (B) Two decimal places
- (C) Four decimal places
- (D) The nearest tenth

When should you use three decimal places?

- (A) For all values
- (B) For very small p-values (e.g., $p = .002$) or when more precision is needed
- (C) Never
- (D) Only for correlations

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3 Part 2: Interpretation - Reporting t-Tests

3.1 Question 7: Independent t-Test Format

The correct APA format for an independent t-test is:

- (A) $t = 2.45$, $p = .018$

- (B) t(48) = 2.45, p = .018, d = 0.71
- (C) t-test: 2.45, significant
- (D) t(n=50) = 2.45, sig = .018

3.2 Question 8: Paired t-Test Format

A paired t-test with 25 participants would be reported as:

- (A) t(25) = 3.12, p = .004
- (B) t(24) = 3.12, p = .004, d = 0.62
- (C) t = 3.12, df = 25, p = .004
- (D) paired t = 3.12

Why df = 24 for n = 25 in a paired test?

- (A) It's a mistake
- (B) For paired tests, df = n - 1
- (C) Half the sample is in each condition
- (D) One participant was excluded

3.3 Question 9: Negative t-Values

When reporting t(28) = –2.89, should the negative sign be included?

- (A) No, always report absolute values
- (B) Yes, it indicates the direction of the effect
- (C) Only if specifically asked
- (D) Negative values are errors

3.4 Question 10: Complete t-Test Reporting

Which is the most complete and correct report?

```
opts_t <- c(
  "The t-test was significant (p < .05)",
  "t = 2.45, p = .018",
  answer = "An independent samples t-test revealed a significant difference,
t(48) = 2.45, p = .018, d = 0.71",
  "There was a significant effect of condition on the DV"
)
```

- (A) The t-test was significant (p < .05)
- (B) t = 2.45, p = .018
- (C) An independent samples t-test revealed a significant difference, t(48) = 2.45, p = .018, d = 0.71
- (D) There was a significant effect of condition on the DV

4 Part 3: Interpretation - Reporting Correlations

4.1 Question 11: Correlation Format

The correct APA format for a correlation is:

- (A) r = .45, significant
- (B) r(48) = .45, p = .001
- (C) correlation = 0.45, p-value = 0.001
- (D) Pearson r: .45*

4.2 Question 12: Degrees of Freedom for Correlations

For a correlation with n = 50, df =

- (A) 50
- (B) 49
- (C) 48
- (D) 100

The formula is:

- (A) df = n
- (B) df = n - 1
- (C) df = n - 2
- (D) df = n/2

4.3 Question 13: Reporting Correlation Strength

When describing r = -.62 in text, you might write:

- (A) A weak negative correlation
- (B) A moderate-to-strong negative correlation
- (C) A strong positive correlation
- (D) No correlation

4.4 Question 14: R-Squared Reporting

If you report r2, how many decimal places?

- (A) 0
- (B) 1
- (C) 2 (e.g., r2 = .36)
- (D) 4

Reporting both r and r2:

- (A) Is redundant and unnecessary
- (B) Can be useful as r2 shows proportion of variance explained
- (C) Is required in all cases
- (D) Is incorrect

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5 Part 3: Interpretation - Reporting Descriptive Statistics

5.1 Question 15: Means and SDs

The correct format for reporting a mean and standard deviation is:

- (A) mean = 45.30, std = 8.25
- (B) Average: 45.30 (8.25)
- (C) M = 45.30, SD = 8.25
- (D) 45.30 +/- 8.25

5.2 Question 16: Reporting Groups

When comparing two groups, descriptive statistics should:

- (A) Only be in a table
- (B) Be reported for each group, either in text or in a table
- (C) Only include the combined mean
- (D) Be omitted if there's a significant difference

Example: "The experimental group (*M* = 72.40, *SD* = 8.50) scored higher than the control group (*M* = 65.20, *SD* = 9.10)."

Is this correct? TRUE / FALSE

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6 Part 4: Application Questions

6.1 Question 17: Correcting Errors

Identify what is WRONG with each report:

"t(48) = 2.45, p = 0.018" - Error:

- (A) Wrong degrees of freedom
- (B) Leading zero on p-value
- (C) Missing effect size
- (D) All of these

"p = .000" - Error:

- (A) Wrong symbol
- (B) Should be p < .001 (p cannot equal exactly zero)
- (C) Too many zeros
- (D) Needs italics

"The correlation was r = .45 (p = significant)" - Error:

- (A) Correlation is wrong
- (B) Should report exact p-value, not just 'significant'
- (C) r should be italicised
- (D) Multiple errors

6.2 Question 18: Writing a Complete Results Sentence

Fill in the blanks for correct APA format:

"An independent samples t-test revealed that participants in the music condition (

- (A) M = 72.40, SD = 8.50
- (B) mean = 72.40, std = 8.50
- (C) 72.40 (8.50)
- (D) M: 72.40

) recalled significantly more words than those in the silence condition (

- (A) M = 65.20, SD = 9.10
- (B) mean = 65.20
- (C) 65.20 average
- (D) M = 65.20 standard deviation = 9.10

),

- (A) t = 2.89, p = .006
- (B) t(48) = 2.89, p = .006, d = 0.82
- (C) t-test significant
- (D) t(48) = 2.89, significance = .006

."

6.3 Question 19: Results vs Discussion Content

Which belongs in the Results section?

- (A) t(48) = 2.89, p = .006, d = 0.82
- (B) This finding supports Smith's (2020) theory
- (C) The study had several limitations
- (D) Future research should examine...

Which belongs in the Discussion section?

- (A) M = 45.30, SD = 8.25
- (B) These results are consistent with previous research showing...
- (C) r(48) = .45, p = .001
- (D) The experimental group scored higher

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7 Part 5: MCQ Exam Practice

7.1 Scenario 1: Report the t-Test

Independent t-test results: Group A (n=25): M=78.4, SD=10.2; Group B (n=25): M=72.1, SD=11.5; t=2.05, p=.046, d=0.58

1. **Degrees of freedom:**

- (A) 50
- (B) 48
- (C) 25
- (D) 24

2. **Correct p-value format:**

- (A) p = 0.046
- (B) p = .046
- (C) p < .05
- (D) sig = .046

3. **Effect size is:**

- (A) Small
- (B) Medium
- (C) Large
- (D) Not reported

4. **Complete correct report:**

```
opts_report <- c(
  "t = 2.05, p = .046",
  "t(50) = 2.05, p = 0.046, d = 0.58",
  answer = "t(48) = 2.05, p = .046, d = 0.58",
  "The groups were significantly different"
)
```

- (A) t = 2.05, p = .046
- (B) t(50) = 2.05, p = 0.046, d = 0.58
- (C) t(48) = 2.05, p = .046, d = 0.58
- (D) The groups were significantly different

7.2 Scenario 2: Report the Correlation

Correlation results: n=60, r=.42, p=.0008

1. **Degrees of freedom:**

- (A) 60
- (B) 59
- (C) 58
- (D) 42

2. **How should p be reported?**

- (A) p = .0008
- (B) p = 0.001
- (C) p < .001
- (D) p = .001

3. **Correct report:**

- (A) r = .42, p < .001
- (B) r(58) = .42, p < .001
- (C) correlation = .42, significant
- (D) r(60) = .42, p = .0008

4. **Describing the relationship:**

- (A) Weak
- (B) Moderate positive
- (C) Strong
- (D) Negative

7.3 Scenario 3: Spot the Errors

A student writes: "We found a significant effect (t = 2.34, p = 0.023). The mean for Group 1 was 56.7 and Group 2 was 48.2."

Identify the errors:

1. **Missing from t-test report:**

- (A) Degrees of freedom and effect size
- (B) The p-value
- (C) Significance statement
- (D) Group labels

2. **p-value format error:**

- (A) Should not have leading zero (should be p = .023)
- (B) Should be p < .05
- (C) Should be rounded
- (D) No error

3. **Descriptive statistics error:**

- (A) Missing SDs and proper formatting (M = , SD =)
- (B) Means are wrong
- (C) Should only report one group
- (D) Numbers should be whole

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8 Summary Checklist

Before writing your Results section, verify:

- ☐ Statistical symbols are italicised (*t*, *p*, *r*, *M*, *SD*, *d*)
- ☐ Degrees of freedom in parentheses after test statistic
- ☐ No leading zeros for values that cannot exceed 1 (*p*, *r*)
- ☐ Two decimal places for most values

- ☐ $p < .001$ for very small p-values (never $p = .000$)
- ☐ Effect sizes reported
- ☐ Descriptive statistics (M, SD) for each group/condition
- ☐ Results reported without extensive interpretation

! Common Mistakes to Avoid

1. Writing "p = 0.023" instead of "p = .023"
2. Writing "p = .000" instead of "p < .001"
3. Forgetting degrees of freedom: "t = 2.45" instead of "t(48) = 2.45"
4. Omitting effect sizes
5. Not italicising statistical symbols